THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — MATHS · NCERT-TYPE CHAPTER TEST SET 3

Chapter 8: **Introduction to Trigonometry** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If $\sec\theta = \frac{17}{8}$ then $\cos\theta =$ 1

(a) $\frac{8}{17}$ **(b)** $\frac{15}{8}$ **(c)** $\frac{17}{8}$ **(d)** $\frac{15}{17}$

2. If $\tan A = \frac{24}{7}$, then $\sin A =$ 1

(a) $\frac{24}{25}$ **(b)** $\frac{25}{24}$ **(c)** $\frac{24}{7}$ **(d)** $\frac{7}{25}$

3. If $\cos A = \frac{20}{29}$, then $\tan A =$ 1

(a) $\frac{21}{29}$ **(b)** $\frac{20}{21}$ **(c)** $\frac{29}{20}$ **(d)** $\frac{21}{20}$

4. The value of $2\tan^2 45^\circ + \cos^2 60^\circ$ is: 1

(a) $\frac{9}{4}$ **(b)** $\frac{13}{4}$ **(c)** 2 **(d)** $\frac{9}{2}$

5. $\tan 65^\circ \cdot \tan 25^\circ =$ 1

(a) 65 **(b)** 2 **(c)** 1 **(d)** 0

6. The value of $\sin 48^\circ / \cos 42^\circ$ is: 1

(a) 0 **(b)** 1 **(c)** 48 **(d)** 2

7. $(5\sin\theta)^2 + (5\cos\theta)^2 =$ 1

(a) 1 **(b)** 25 **(c)** 10 **(d)** 5

8. If $\tan A = \frac{21}{20}$, then $\sin A =$ 1

(a) $\frac{20}{29}$ **(b)** $\frac{29}{21}$ **(c)** $\frac{21}{20}$ **(d)** $\frac{21}{29}$

9. If $\cos A = \frac{9}{41}$, then $\tan A =$ 1

(a) $\frac{9}{40}$ **(b)** $\frac{40}{9}$ **(c)** $\frac{41}{9}$ **(d)** $\frac{40}{41}$

10. The value of $\sin^2 60^\circ + \cos^2 60^\circ$ is: 1

(a) 2 **(b)** 1 **(c)** $\frac{3}{2}$ **(d)** $\frac{3}{4}$

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Evaluate: $\dfrac{\tan 47^\circ}{\cot 43^\circ}$. 2

12. If $\tan A = \frac{24}{7}$, find $\sec A$. 2

13. Evaluate: $\dfrac{\sin 14^\circ}{\cos 76^\circ} + \dfrac{\tan 61^\circ}{\cot 29^\circ}$. 2

14. Evaluate: $\dfrac{\tan 58^\circ}{\cot 32^\circ}$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. If $\tan\theta = \dfrac{40}{9}$, find $\sin\theta$, $\cos\theta$ and $\sec\theta$. 3

16. In $\triangle ABC$, right-angled at B, $AB = 9$ cm and $BC = 12$ cm. Find $\sin A$, $\cos A$ and $\tan A$. 3

17. Prove that: $\dfrac{1-\tan^2 A}{1+\tan^2 A} = \cos^2 A - \sin^2 A$. 3

OR

If $\tan\theta = \dfrac{8}{6}$, find $\sin\theta$, $\cos\theta$ and $\sec\theta$. 3

18. In $\triangle ABC$, right-angled at B, $AB = 12$ cm and $BC = 16$ cm. Find $\sin A$, $\cos A$ and $\tan A$. 3

OR

Prove that: $(1+\tan^2 A)\sin^2 A = \tan^2 A$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Prove that: $(1+\cot A - \operatorname{cosec} A)(1+\tan A+\sec A) = 2$. 5

OR

If $\tan A = \dfrac{21}{20}$, evaluate $\dfrac{5\sin A + 4\cos A}{5\sin A + 2\cos A}$. 5

20. Prove that: $\sec A(1-\sin A)(\sec A+\tan A) = 1$. 5

OR

If $\tan A = \dfrac{40}{9}$, evaluate $\dfrac{7\sin A + 2\cos A}{4\sin A + 1\cos A}$. 5

END OF QUESTION PAPER — Total: 40 Marks

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